Elita Lestari Setianto (201344008)	**Rangkuman Minggu 5**	Selasa, 27 September 2022
		16TNK5043 (Aplikasi Mikrokontroler)
3NK		Ferry Satria, BSEE.,M.T. Rahmawati Hasanah,S.ST.,M.T.

Mengenable semua pake SEI

Mengdisable semua pake CLI

Klo mau membuat

int 1 pake bit 7

Int 0 bit 6

Int 2 bit 5

Enable 0 pake logic 1 (GICR)

Misal int 1 mau di enable jadi 10000000

LDI R20, 0X80

Out gicr,r20

Klo mau int1 jadi rising edge (MCUCR)

LDI R20, 0X0C

OUT MCUCR,R20

Table 35. Interrupt 1 Sense Control

ISC11	ISC10	Description
0	0	The low level of INT1 generates an interrupt request.
0	1	Any logical change on INT1 generates an interrupt request.
1	0	The falling edge of INT1 generates an interrupt request.
1	1	The rising edge of INT1 generates an interrupt request.

Klo mau int 0 jadi low level

LDI R20,0

OUT MCUCR,R20

Table 36. Interrupt 0 Sense Control

ISC01	ISC00	Description
0	0	The low level of INT0 generates an interrupt request.
0	1	Any logical change on INT0 generates an interrupt request.
1	0	The falling edge of INT0 generates an interrupt request.
1	1	The rising edge of INT0 generates an interrupt request.

INTERRUPT VEKTOR

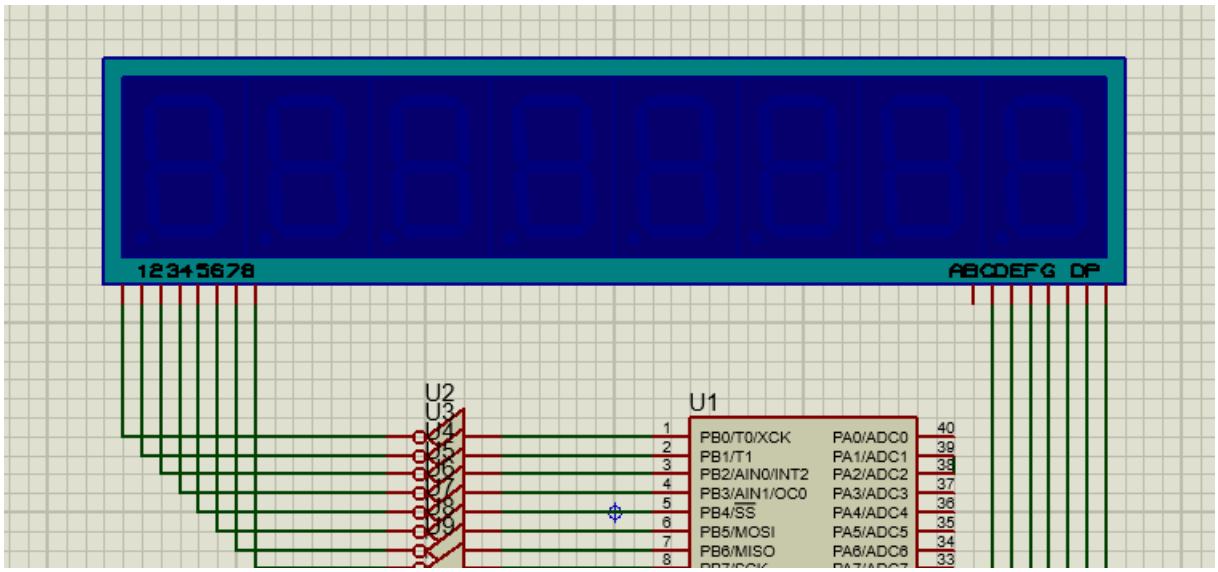
INT 0 : 001

INT 1 : 010

INT 2 : 011

Perintah untuk mengakhiri interrupt

Klo ada 8 7 segmen harus menghubungkan semua jadi Satu



12345678 :

ABCDEFG : common

Program menampilkan huruf Rajin

```
;Program Interrupt dan Timer
; pakai project "ROOT"
.include "m8535def.inc"
.org 0x00
.equ timer_1=0xFFF0
rjmp main

.org 0x001
rjmp int_0
```

```
main :
        ldi r16,low(ramend)
        out spl,r16
        ldi r16,high(ramend)
        out sph,r16

INT_1:
        cli
        ldi r17,0x20
        out gicr,r17 ; DISABLE SEMUA INTERRUPT

        ldi r17,0x01
        out mcucsr,r17 ;PILIH SISI NAIK ATAU SISI TURUN

        ldi r17,0x40 ; ENABLE GENERAL INT
        out gicr,r17

        ldi r17,0xFF ; ATUR PORTB DAN PORTC SBG OUTPUT
        out ddrb,r17
        out ddrc,r17

        sei    ; SET GENERAL INT
        clr r20 ; R20 COUNTER
;====================
loop:
        ldi   zl,low(2*posisi)
        ldi zh,high(2*posisi)

        ldi r26,$60 ; inisialisasi pointer x
        clr r27

lup:
        lpm r18,z+ ; Data Posisi dipindahkan ke Data Area
        st x+,r18
        cpi r26,$68
        brne lup
```

```
;======================
lup2:
        rcall nyala0
        andi r20,0x01 ; JIKA R20 GENAP, TDK BERKEDIP
        breq lup2         ; JIKA R20 GANJIL, BERKEDIP
        rcall mati0
        rjmp lup2

nyala0:

        cli ; CLEAR INT
        ldi r29,$0F ; R29=15

nyala:
```

```
        ldi r26,$60 ;POINTER X MENUNJUK KE DATA AREA
        clr r27     ; .....
        ldi zl,low(2*pola) ; POINTER Z menunjuk ke POLA KARAKTER
        ldi zh,high(2*pola)

nyala1:
        ld r28,x+
;       com r28
        out portb,r28

        lpm     r21,z+
        out portc,r21

        rcall  delay_2ms
        cpi       r26,$68
        brne nyala1

        dec      r29
        brne  nyala
        sei
        ret
;=========================

;============================
mati0:
        cli
        ldi r29,125

mati:
        ldi  r21,0xFF
        out portc,r21

        rcall delay_2ms

        dec     r29
        brne  mati
        sei
        ret
```

```
;============================

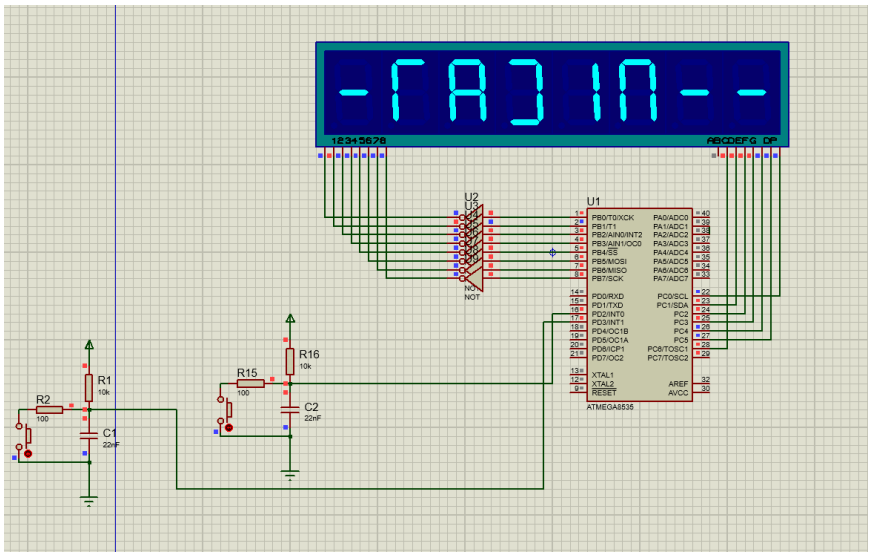
delay_2ms:
        cli
        push r16
        push r17
        ldi r16,4
        out timsk,r16
        ldi   r16,  high(timer_1)
        out tcnt1h,r16
        ldi r16, low(timer_1)
        out tcnt1l,r16
        ldi r16,5
        out tccr1b,r16
```

```
looptimer:
  in r17,tifr
  sbrs r17,tov1
  rjmp looptimer
  ldi r16,4
  out tifr,r16
  pop r17
  pop r16
  sei
  ret

INT_0:
        cli
        inc r20
        sei
        reti

pola:
.db 0xBF, 0xCE, 0x88, 0xF0, 0xF9, 0xC8, 0xBF, 0xBF;karakter apa yg akan dimunculkan

posisi:
.db 0xFE, 0xFD, 0xFB, 0xF7, 0xEF, 0xDF, 0xBF, 0x7F;digit mana yg digunakan, cmn bisa salah satu yg aktif gak bisa 8 8 nya, krn commonnya 1 per satu aktifnya (sistem scan?)
```



Program menampilkan huruf belajar

```
.INCLUDE"M8535DEF.INC"
.ORG 0
RJMP MAIN
                ; BCD UP-DOWN COUNTER MODULO 36, DENGAN 74LS47
                ; DELAY MENGGUNAKAN LOOP 3 REGISTER
          ; IMPLEMENTASI: PROTEUS DEC_COUNT_01 [DI PROTEUS_PROJECT]
```

```
MAIN:LDI R16,LOW(RAMEND)
        OUT SPL,R16
        LDI R16,HIGH(RAMEND)
        OUT SPH,R16

        LDI R20,0
        OUT DDRA,R20      ; PORT A SBG INPUT
        LDI R20,0XFF
        OUT DDRB,R20      ; PORT B SBG OUTPUT
        LDI R21,0X99      ; KOMPLEMEN 10 DARI 1
ITU:  LDI R18,0           ; INPUT AWAL UP COUNTER
SANA:OUT PORTB,R18
        RCALL DELAY
        IN R17,PINA       ; MEMBACA DATA DARI PIN A
        MOV R16,R17
        ANDI  R16,1
        CPI R16,0
        BRNE DOWN              ; JUMP KE DOWN COUNTER JIKA R16 = 1
UP:     LDI R20,1
        MOV  R19,R18
        ADD  R19,R20
        RCALL  DAA
        MOV R18,R19
        CPI  R18,0X36;  BATAS  ATAS  UP  COUNTER
        BRNE SANA
        RJMP ITU



INI:  LDI R18,0X35        ; NILAI AWAL DOWN COUNTER
DOWN:
SINI: OUT PORTB,R18
        RCALL DELAY
        IN R17,PINA       ; MEMBACA DATA DARI PIN A
        MOV R16,R17
        ANDI R16,1
        CPI R16,1
        BRNE UP                ; JUMP KE UP COUNTER JIKA R16 = 0
```

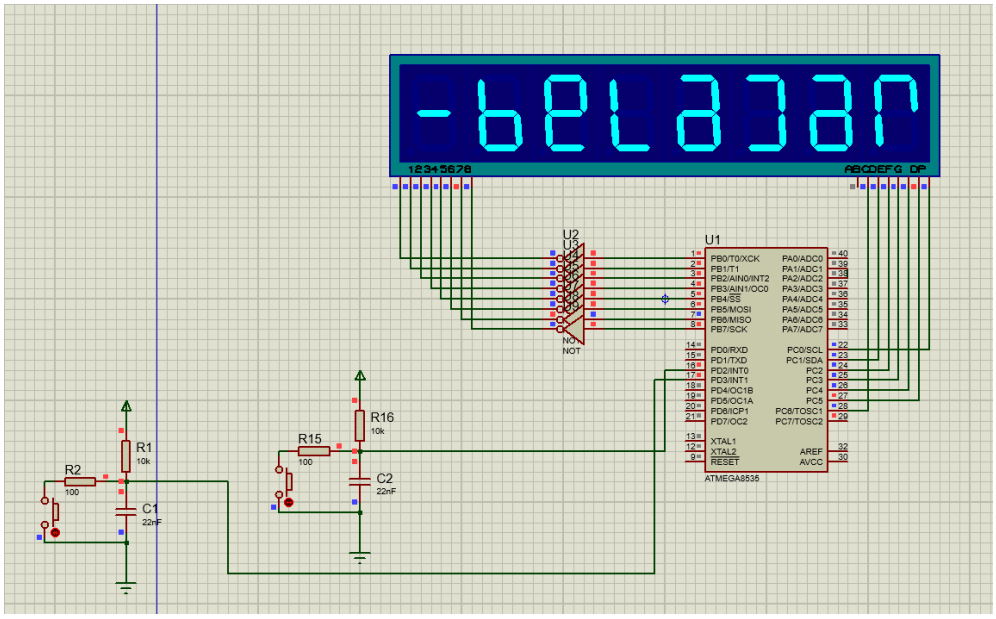
```
                MOV R19,R18
                ADD  R19,R21
                RCALL   DAA
                MOV R18,R19
                CPI R18,0x99 ; BATAS BAWAH DOWN COUNTER
                BRNE SINI
                RJMP INI

DELAY:      LDI R23,3
LUP1:   LDI   R24,0XFF
LUP2: LDI R25,0
LUP3: DEC R25
                BRNE LUP3
                DEC      R24
                BRNE LUP2
```

```
                DEC     R23
                BRNE LUP1
                RET
daa: push r16;amankan isi reg. ke stack
                push r17
                push r21
                push r24
                in r16,sreg; amankan sreg
                in r17,sreg
                ldi r24,0 ; r24 hanya sbg status dr carry flag
                adc r24,r24; status carry flag sebelum koreksi
; ambil, amankan kembali sreg, uji half carry
                out sreg,r17
                brhc lup4; lompat ke lup4 jika H=0
                rcall fk_ln; satuan + 06 jika H=1
        ;       rjmp lup5
lup4:   mov r21,r19
                andi r21,$0f; ambil dgt.satuan
                cpi r21,$0a ; > 9
                brcs lup5; lompat ke lup5 jika < 9
                rcall fk_ln; satuan + 06 jika > 9
lup5:   out sreg,r17; ambil, amankan kembali sreg, uji carry
                brcc lup6; lompat ke lup6 jika C=0
                rcall fk_hn; puluhan + 60 jika C=1
        ;       rjmp lup7
lup6:   mov r21,r19
                andi r21,0xf0; ambil dgt.puluhan
                cpi r21,0xa0
                brcs lup7; lompat ke lup7 jika pul < a
                rcall fk_hn; pul + 60 jika > 9
lup7:   out sreg,r16; mengatur carry flag sblm keluar:
                clc; C = 0
                in r16,sreg ;amankan sreg stlh c=0
                cpi r24,0 ;menguji status
                breq lup8
                out sreg,r16;ambil sreg, set c
                sec; C = 1
```

```
                in r16,sreg; amankan sreg stlh c=1
lup8:
                out sreg,r16;status akhir
                pop r24;kembalikan isi reg. dr stack
                pop r21
                pop r17
                pop r16
                ret
fk_ln: ldi r21,06; koreksi dgt.satuan
                out sreg,r16
                add  r19,r21
                in   r16,sreg
                brcc tuh
                ldi  r21,0x01
                add r17,r21
```

```
tuh: out sreg,r16
                adc  r24,r24
                ret
fk_hn: ldi r21,$60; koreksi dgt.puluhan
                out sreg,r16
                add  r19,r21
                in r16,sreg
                adc r24,r24; amankan carry pd r24
                ret ; akhir dari prog. subrutin.
```



Buat program untuk menampilkan huruf A pada unit peraga 7 Segmen common Anoda, jika tombol int 0 ditekan akan menyebabkan : tampil a kecil selama 2 detik saja, pake push button untuk mengaktifkan int0.

```
.INCLUDE"M8535DEF.INC"

.ORG 0 RJMP
MAIN

.ORG 1 RJMP
INT_0


MAIN:LDI R16,LOW(RAMEND)
                OUT SPL,R16
                LDI R16,HIGH(RAMEND)
                LDI R18,8                       ; A KAPITAL PADA 7 SEGMENT CA
                OUT PORTA,R18

AKHIR: RJMP AKHIR
```

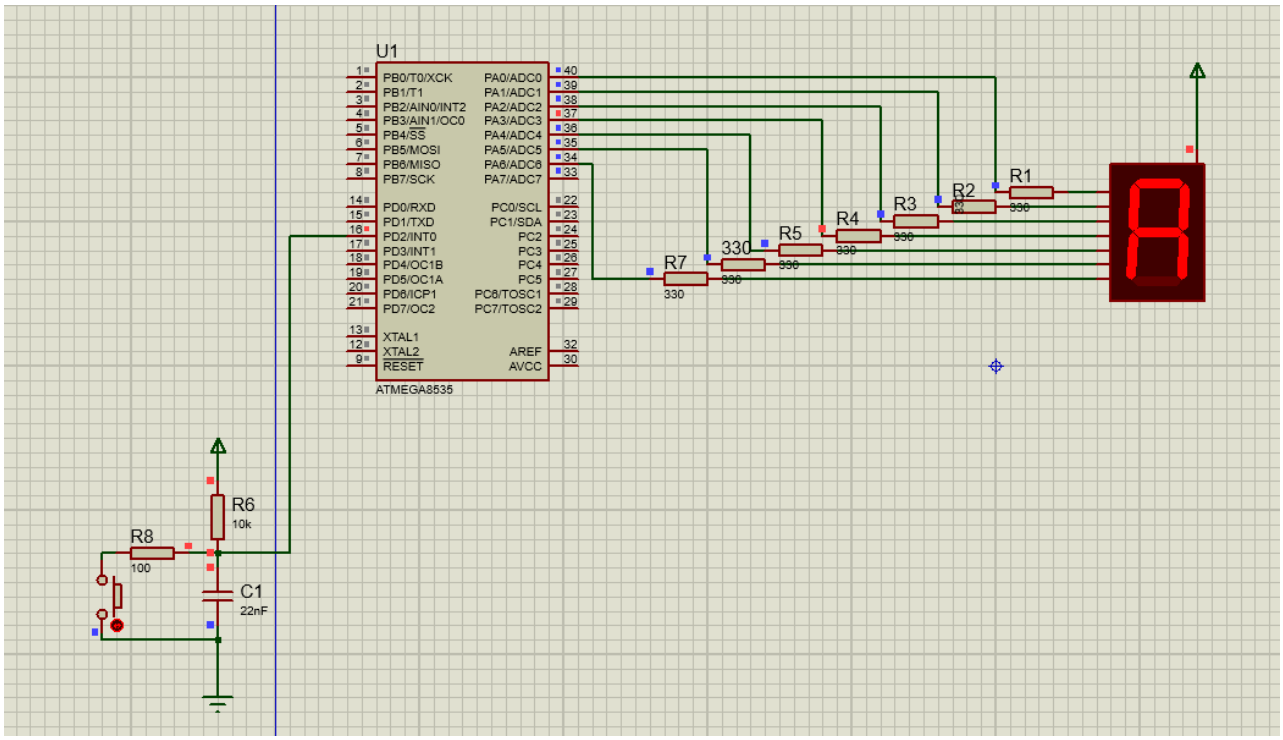
```
DELAY:          CLI
                LDI R22,3
LUP1: LDI R24,0XFF
LUP2: LDI R25,0 LUP3:
DEC R25

                BRNE LUP3
                DEC R24
                BRNE LUP2
                DEC R23
                BRNE LUP1


INT_0:CLI
                LDI R19,0X40
                OUT GICR,R19            ; GICR HANYA MENGENABLE INT0
                LDI R20,3

                LDI R26,0X20 OUT        ; A KECIL PADA 7 SEGMENT CA
                PORTA,R26 RCALL
                DELAY RCALL
                DELAY

                SEI
                RETI
```



Soal

1. Rancang suatu up counter heksadesimal 1 digit modulo 16 menghitung (0 – F) secara berulang, gunakan 7 segment common katoda. Kenaikan nilai counter dikendalikan melalui tombol interrupt 0 (sebagai increment).

   Cara menyelesaikan :
   Kata kunci
   Up counter hexadecimal 1 digit (0-F) menggunakan push button.
   7 segment common katoda → terhubung ke ground, dan diberi logic 1 untuk ON, logic 0 untuk OFF, menggunakan PORT A

   POLA 7 SEGMENT

   0 → 3F
   1 →
   2 →
   3 →
   4 →
   5 →
   6 →
   7 → 07
   8 →
   9

   Tugas Program Utama → lakukan inisialisasi : STACK, GICR = 0X40, MCUCR = 2 juga bertugas untuk menentukan isi counter

   *counter harus di inisialisasi,

   Tugas Program Interrupt → Naikkan counter 0 – F
   SEI → Setelah proses inisialisasi beres

- Kode Program

```
.INCLUDE"M8535DEF.INC"
.ORG 0
RJMP MAIN

.ORG 1
RJMP INT_0

MAIN:LDI R16,LOW(RAMEND)
        OUT SPL,R16
        LDI R16,HIGH(RAMEND)
        OUT SPH,R16

        CLI
        LDI R19,0X40
        OUT GICR,R19          ; REGISTER GICR HANYA
MENGENABLE INT0
        LDI R20,2
        OUT MCUCR,R20         ; INT0 AKTIF SAAT
FALLING EDGE (sisi turun)
        SEI



        LDI R17,0XFF
        OUT DDRA,R17          ; PORT B SEBAGAI OUTPUT

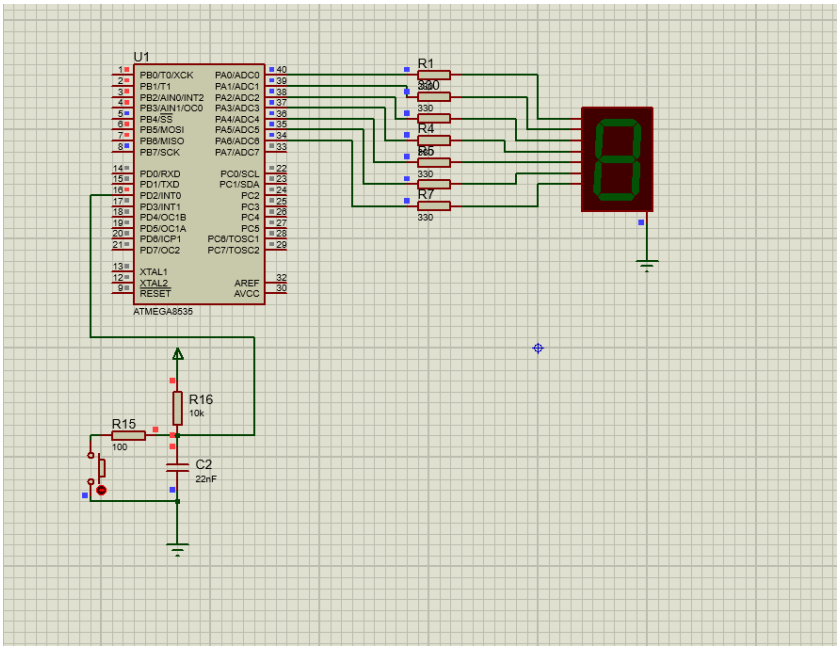
        ldi zl,low(2*HEX)     ; POINTER Z menunjuk ke HEX
KARAKTER
        ldi zh,high(2*HEX)
        LD R26,Z+
        OUT PORTA,R18
```

```
        DELAY:      CLI
                    LDI  R23,3
        LUP1: LDI  R24,0XFF
        LUP2: LDI  R25,0
        LUP3: DEC  R25
                    BRNE LUP3
                    DEC  R24
                    BRNE LUP2
                    DEC  R23
                    BRNE LUP1
                    SEI
                    RET

        INT_0:CLI

                    OUT   PORTA,R26
                    RCALL DELAY
                    RCALL DELAY
                    SEI
                    RETI

        HEX:
.db 0X3F
```



2. Rancang suatu DOWN counter heksadesimal 1 digit modulo 16 menghitung (0 – F) secara berulang, gunakan 7 segment common katoda. Kenaikan nilai counter dikendalikan melalui tombol interrupt 0 (sebagai increment).

```
.include "m8535def.inc"
.org 0
RJMP MAIN

.ORG 0X01 RJMP INT_0

.ORG 0X02 RJMP INT_1
;Rancang suatu Up/Down Counter Heksadesimal 1 digit Modulo 16 (Menghitung naik/turun
;secara berulang dengan range output dari 0 sampai F). Gunakan seven segment jenis
;Common Cathode dengan tombol interupsi INT0 yang berfungsi untuk menaikkan/menurunkan
;nilai counter (Counter incrementer/decrementer); dan INT1 yang berfungsi sebagai
;pengatur Mode kerja Counter (up/down).
```

MAIN:CLI

```
LDI R16,LOW(RAMEND) OUT SPL,R16
LDI R16,HIGH(RAMEND) OUT SPH,R16

LDI R20,0XFF OUT DDRA,R20

LDI R20,0XC0 OUT GICR,R20

LDI R20,0X0F OUT MCUCR,R20

LDI R30,LOW(2*POLA) ; INIT POINTER Z LDI R31,HIGH(2*POLA)
LDI R22,0 ; INIT COUNTER SEI


KK:     MOV R30,R22 ;MENAMPILKAN ISI COUNTER LPM R2,Z
OUT PORTA,R2 RCALL DELAY RJMP KK

DELAY:        CLI
LDI R23,1 LP1:        LDI R24,4
LP2:    LDI R25,0 LP3:        DEC R25
BRNE LP3 DEC R24 BRNE LP2 DEC R23 BRNE LP1 SEI
RET
;================


INT_0:CLI

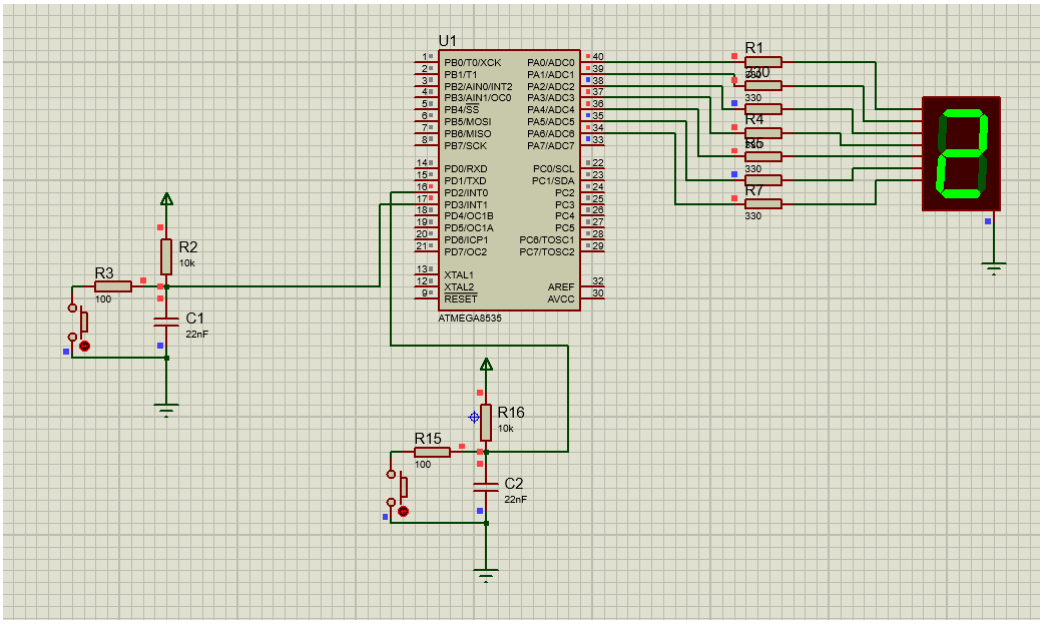

MOV R27,R20 ANDI R27,1 CPI R27,1 BRNE TRN INC R22
CPI R22,0X10 BRNE LK1 LDI R22,0 RJMP LK1

TRN: DEC R22
CPI R22,0XFF BRNE LK1 LDI R22,0X0F

LK1:; SEI
```

RETI

INT_1:CLI
INC R20 SEI RETI
;==================
.ORG 0X80 POLA:
.DB 0X3F,0X06,0X5B,0X4F,0X66,0X6D,0X7D,0X07
.DB 0X7F,0X6F,0X77,0X7C,0X39,0X5E,0X79,0X71



3. Rancang suatu UPDOWN counter heksadesimal 1 digit modulo 16 menghitung (0 – F) secara berulang, gunakan 7 segment common katoda. Kenaikan nilai counter dikendalikan melalui tombol interrupt 0 (sebagai increment).
   INT 1 = GANTI MODE
   INT 0 = MODE KERJA

- Kode Program

```
.include "m8535def.inc"
.org 0
RJMP MAIN


.ORG 0X01
RJMP INT_0
```

```
.ORG 0X02
RJMP INT_1
; Up-Counter Heksadesimal Modulo 16 (Menghitung secara berulang
; dari 0 sampai F); INT0 berfungsi sebagai Counter Incrementer
; Menggunakan Seven Segment Common Cathode.
; PROTEUS PROJECT: UP-CNT-INT0


MAIN:CLI
        LDI R20,LOW(RAMEND)
        OUT SPL,R20
        LDI R20,HIGH(RAMEND)
        OUT SPH,R20
        LDI R20,0XFF
        OUT DDRA,R20
        LDI R20,0XC0
        OUT GICR,R20
        LDI R20,0X0F
        OUT MCUCR,R20
        LDI R30,LOW(2*POLA) ; INIT POINTER Z
        LDI R31,HIGH(2*POLA)
        LDI R22,0 ; INIT COUNTER
        SEI
KK:     MOV R30,R22
        LPM R2,Z
        OUT PORTA,R2
        RCALL DELAY
        RJMP KK

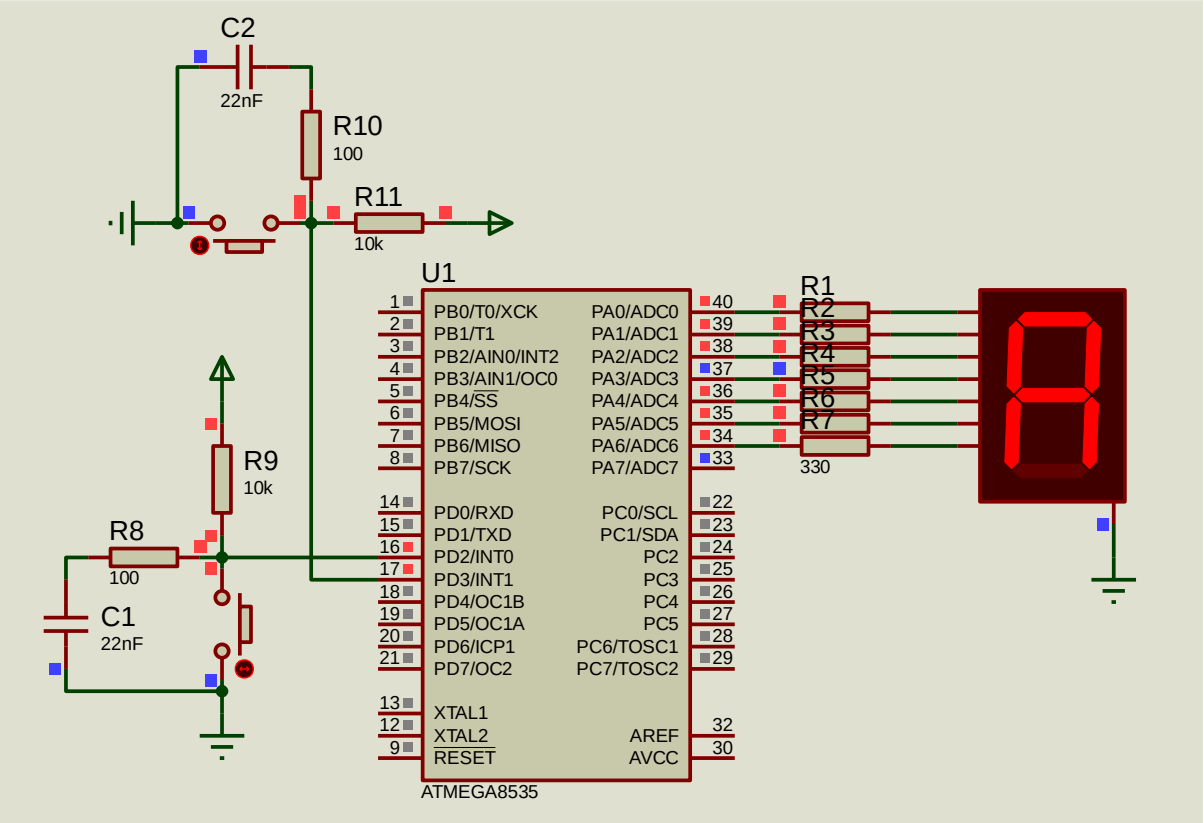
DELAY:  CLI
        LDI R23,1
LP1:    LDI R24,4
```

```
LP2:    LDI R25,0
LP3:    DEC R25
                BRNE LP3
                DEC R24
                BRNE LP2
                DEC R23
                BRNE LP1
                SEI
                RET
;=================
INT_0:CLI
                MOV R21,R20
                ANDI R21,1
                CPI R21,1
                BRNE TRN
                INC R22
                CPI R22,0X10
                BRNE LK1
                LDI R22,0
                RJMP LK1
TRN:    DEC R22
                CPI R22,0XFF
                BRNE LK1
                LDI R22,0X0F
LK1:    SEI
                RETI

INT_1:CLI
                INC R20
                SEI
                RETI
;=================
.ORG 0X80
```

```
        POLA:
        .DB 0X3F,0X06,0X5B,0X4F,0X66,0X6D,0X7D,0X07
.DB 0X7F,0X6F,0X77,0X7C,0X39,0X5E,0X79,0X71
```

- Hasil Program



4. Rancang suatu Up Counter Heksadesimal 2 digit Modulo 45 (Menghitung naik secara berulang dari 00 sampai 2C). Gunakan seven segment jenis Common Anode dengan tombol interupsi INT2 yang berfungsi untuk menaikkan nilai counter (Counter incrementer).

- Kode Program

```
        .include "m8535def.inc"
        .org 0
        RJMP MAIN


        .ORG 0X012
        RJMP INT_2
```

```
; Up-Counter Heksadesimal Modulo 45 (Menghitung secara berulang
; dari 0 sampai 2C); INT0 berfungsi sebagai Counter Incrementer
; Menggunakan Seven Segment Common Cathode.
; PROTEUS PROJECT: UP-CNT-INT0
;

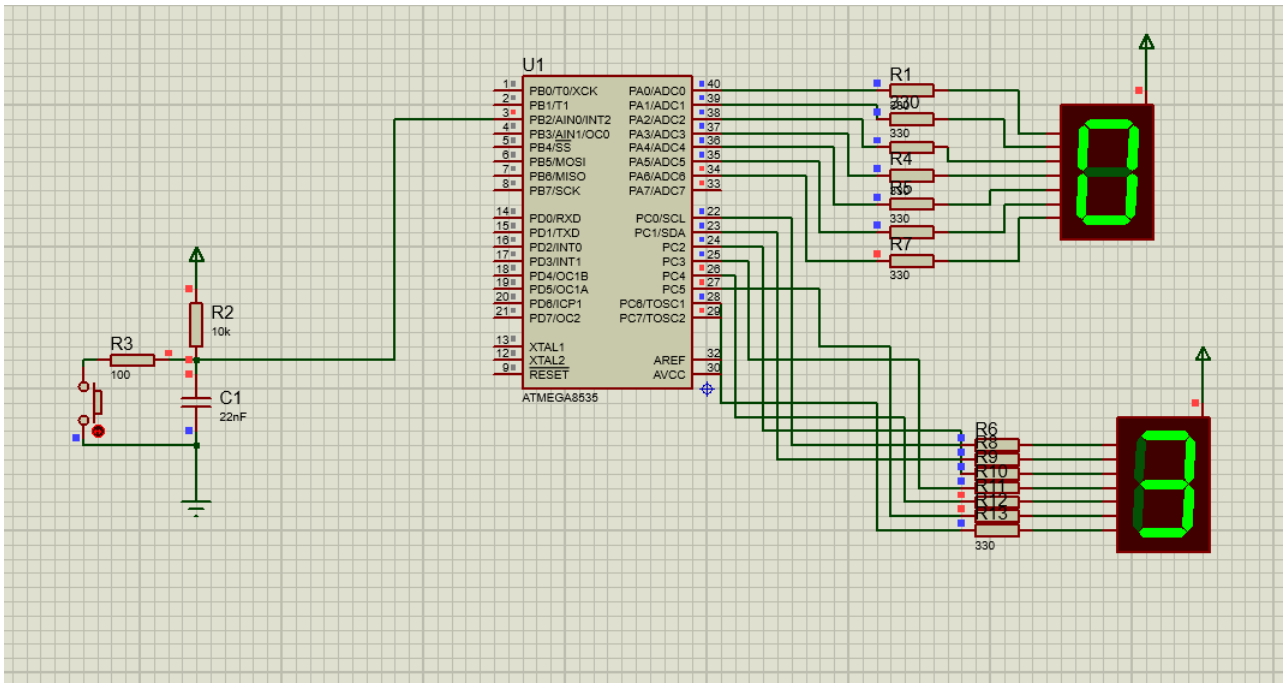
MAIN:CLI
        LDI R20,LOW(RAMEND)
        OUT SPL,R20
        LDI R20,HIGH(RAMEND)
        OUT SPH,R20
        LDI R20,0XFF
        OUT DDRA,R20
        OUT DDRC,R20
        LDI R20,0X20
        OUT GICR,R20
        LDI R20,0X40
        OUT MCUCSR,R20
        LDI R30,LOW(2*POLA) ; INIT POINTER Z
        LDI R31,HIGH(2*POLA)
        LDI R22,0               ; INIT COUNTER
        SEI
KK:     MOV R27,R22
        ANDI R27,0XF0           ;           MENGAMBIL
ENAMBELASAN
        LSR R27
        LSR R27
        LSR R27
        LSR R27
        MOV R30,R27
        LPM R2,Z
        OUT PORTA,R2
        MOV R27,R22
```

```
                ANDI R27,0X0F           ; MENGAMBIL SATUAN
                MOV R30,R27
                LPM R2,Z
                OUT PORTC,R2
                RCALL DELAY
                RJMP KK



DELAY:          CLI
                LDI R23,1
LP1:    LDI R24,4
LP2:    LDI R25,0
LP3:    DEC R25
                BRNE LP3
                DEC R24
                BRNE LP2
                DEC R23
                BRNE LP1
                SEI
                RET
                ;================
INT_2:CLI
                INC R22
                CPI R22,0X2D
                BRNE LK1
                LDI R22,0
LK1:    SEI
                RETI
;=================
.ORG 0X80
POLA:
.DB 0XC0,0XF9,0XA4,0XB0,0X99,0X92,0X82,0XF8
.DB 0X80,0X90,0X88,0X83,0XC6,0XA1,0X86,0X8E
```

- Hasil Program



5. Seperti soal nomor 4, tapi untuk Down Counter.

- Kode Program

```
.include "m8535def.inc"
.org 0
RJMP MAIN

.ORG 0X012
RJMP INT_2
; Up-Counter Heksadesimal Modulo 45 (Menghitung secara berulang
; dari 0 sampai 2C); INT0 berfungsi sebagai Counter Incrementer
; Menggunakan Seven Segment Common Cathode.
; PROTEUS PROJECT: UP-CNT-INT0
;

MAIN:CLI
        LDI R20,LOW(RAMEND)
        OUT SPL,R20
        LDI R20,HIGH(RAMEND)
```

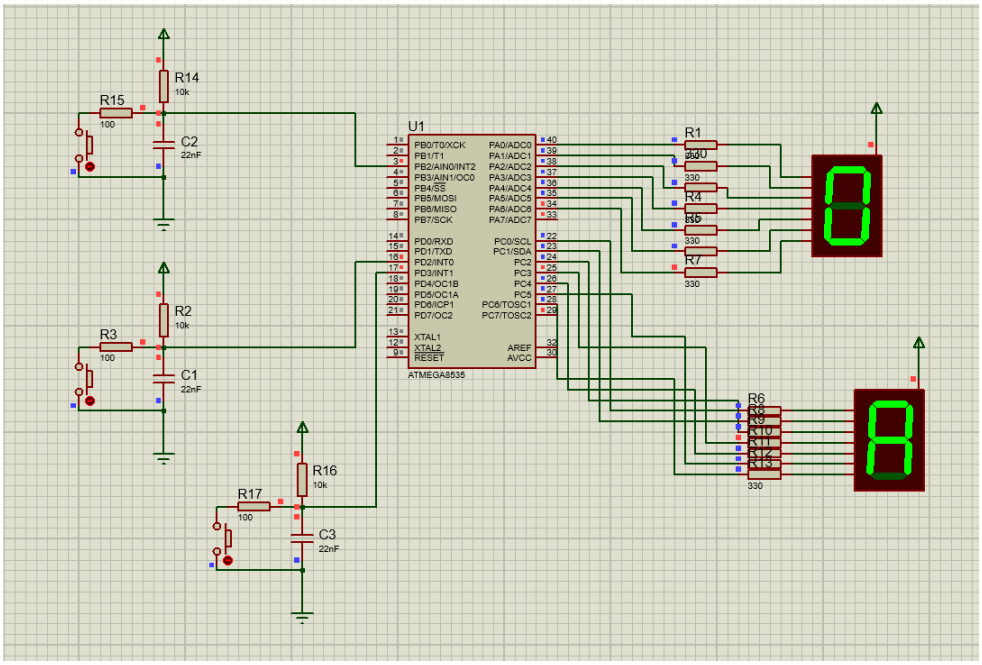
```
                OUT SPH,R20
                LDI R20,0XFF
                OUT DDRA,R20
                OUT DDRC,R20
                LDI R20,0X20
                OUT GICR,R20
                LDI R20,0X40
                OUT MCUCSR,R20
                LDI R30,LOW(2*POLA) ; INIT POINTER Z
                LDI R31,HIGH(2*POLA)
                LDI R22,0X2C            ; INIT COUNTER
                SEI
KK:             MOV R27,R22
                ANDI R27,0XF0           ;           MENGAMBIL
ENAMBELASAN
                LSR R27
                LSR R27
                LSR R27
                LSR R27
                MOV R30,R27
                LPM R2,Z
                OUT PORTA,R2
                MOV R27,R22
                ANDI R27,0X0F           ; MENGAMBIL SATUAN
                MOV R30,R27
                LPM R2,Z
                OUT PORTC,R2
                RCALL DELAY
                RJMP KK



DELAY:          CLI
                LDI R23,1
```

```
        LP1:    LDI R24,4
        LP2:    LDI R25,0
        LP3:    DEC R25
                        BRNE LP3
                        DEC R24
                        BRNE LP2
                        DEC R23
                        BRNE LP1
                        SEI
                        RET
                        ;=================
        INT_2:CLI
                        DEC R22
                        CPI R22,0XFF
                        BRNE LK1
                        LDI R22,0X2C
        LK1:    SEI
                        RETI
        ;==================
        .ORG 0X80
        POLA:
        .DB 0XC0,0XF9,0XA4,0XB0,0X99,0X92,0X82,0XF8
.DB 0X80,0X90,0X88,0X83,0XC6,0XA1,0X86,0X8E
```

- Hasil Program



6. Up/Down-Counter Heksadesimal Modulo 21 dan 35 (Menghitung secara berulang dari 0 sampai F); INT0 berfungsi sebagai Counter Incrementer Menggunakan Seven Segment Common Cathode. INT1 mengubah modulo. INT2 mode kerja counter

- Kode Program

```
include "m8535def.inc"
.org 0
RJMP MAIN

.ORG 0X001
RJMP INT_0

.ORG 0X002
RJMP INT_1

.ORG 0X012
RJMP INT_2
; Up/Down-Counter Heksadesimal Modulo 21 dan 35
; INT0 berfungsi sebagai Counter Incrementer
; Menggunakan Seven Segment Common Anode INT1 mengubah modulo.
INT2 mode kerja counter
```

```
; PROTEUS PROJECT: UP-CNT-INT0



MAIN:CLI
        LDI R20,LOW(RAMEND)
        OUT SPL,R20
        LDI R20,HIGH(RAMEND)
        OUT SPH,R20
        LDI R20,0XFF
        OUT DDRA,R20
        OUT DDRC,R20
        LDI R20,0XE0
        OUT GICR,R20
        LDI R20,0X0F
        OUT MCUCR,R20
        LDI R20,0X40
        OUT MCUCSR,R20
        LDI R30,LOW(2*POLA) ; INIT POINTER Z
        LDI R31,HIGH(2*POLA)
        LDI R22,0               ; INIT COUNTER
        SEI
KK:     MOV R27,R22
        ANDI R27,0XF0           ;           MENGAMBIL
ENAMBELASAN
        LSR R27
        LSR R27
        LSR R27
        LSR R27
        MOV R30,R27
        LPM R2,Z
        OUT PORTA,R2
        MOV R27,R22
        ANDI R27,0X0F           ; MENGAMBIL SATUAN
```

```
                MOV R30,R27
                LPM R2,Z
                OUT PORTC,R2
                RCALL DELAY
                RJMP KK



DELAY:          CLI
                LDI R23,1
LP1:    LDI R24,4
LP2:    LDI R25,0
LP3:    DEC R25
                BRNE LP3
                DEC R24
                BRNE LP2
                DEC R23
                BRNE LP1
                SEI
                RET
;=================
INT_0:CLI
                MOV R21,R17
                ANDI R21,1
                BRNE TRN
                MOV R19,R18
                ANDI R19,1
                BRNE MOD35H
                INC R22
                CPI R22,0X15
                BRLT LK1
                LDI R22,0
                RJMP LK1
MOD35H:         INC R22
```

```
                CPI R22,0X23
                BRNE LK1
                LDI R22,0
                RJMP LK1
TRN:   MOV R19,R18
                ANDI R19,1
                BRNE MD35H
                DEC R22
                CPI R22,0XFF
                BRNE LK1
                LDI R22,0X14
                RJMP LK1
MD35H:          DEC R22
                CPI R22,0XFF
                BRNE LK1
                LDI R22,0X22
LK1:   SEI
                RETI

INT_1:CLI
                INC R18
                SEI
                RETI

INT_2:CLI
                INC R17
                SEI
                RETI
;=================
.ORG 0X80
POLA:
.DB 0XC0,0XF9,0XA4,0XB0,0X99,0X92,0X82,0XF8
.DB 0X80,0X90,0X88,0X83,0XC6,0XA1,0X86,0X8E
```